

Integration By Parts

Section 8.2

1-29-26

A) Undo Product Rule

$$\int \frac{d}{dx} uv = \int v du + \int u dv$$

$$uv = \int v du + \int u dv$$

$$uv - \int v du = \int u dv$$

Ex: $\int x \cos x \, dx =$

$$u = x \quad dv = \cos x \, dx$$

$$du = dx \quad v = \sin x$$

$$uv = x \sin x$$

$$\int v du = \int \sin x \, dx$$

$$uv - \int v du = \int u dv$$

$$x \sin x - \int \sin x \, dx = \int u dv$$

$$x \sin x + \cos x + C$$

Ex: $\int x e^{2x} \, dx =$

$$u = x \quad dv = e^{2x} \, dx$$

$$du = dx \quad v = \frac{1}{2} e^{2x}$$

$$\frac{x e^{2x}}{2} - \int \frac{1}{2} e^{2x} \, dx$$

$$\frac{x e^{2x}}{2} - \frac{1}{4} e^{2x} + C$$

Ex: $\int x \ln x \, dx$

$$u = \ln x \quad dv = x$$

$$du = \frac{1}{x} \, dx \quad v = \frac{x^2}{2}$$

$$\frac{x^2 \ln x}{2} - \int \frac{x^2}{2} \cdot \frac{1}{x} \, dx =$$

$$\frac{x^2 \ln x}{2} - \int \frac{x}{2} \, dx =$$

$$\frac{x^2 \ln x}{2} - \frac{x^2}{4} + C$$

Ex: $\int \ln x \, dx =$

$$u = \ln x \quad dv = 1 \, dx$$

$$du = \frac{1}{x} \quad v = x$$

$$x \ln x - \int 1 \, dx = x \ln x - x + C$$

Ex: $\int \arctan x \, dx =$

$$u = \arctan x \quad dv = 1 \, dx$$

$$du = \frac{1}{x^2 + 1} \, dx \quad v = x$$

$$x \arctan x - \int \frac{x}{x^2 + 1} \, dx =$$

$$u = x^2 + 1 \quad du = 2x \, dx \quad x \arctan x - \frac{1}{2} \int \frac{1}{u} \, du$$

$$x \arctan x - \frac{1}{2} \ln |u| + C$$

$$x \arctan x - \frac{1}{2} \ln |x^2 + 1| + C$$

Ex: $\int x^2 e^{3x} \, dx =$

$$u = x^2 \quad dv = e^{3x}$$

$$du = 2x \, dx \quad v = \frac{1}{3} e^{3x}$$

$$\frac{x^2 e^{3x}}{3} - \frac{2}{3} \int x e^{3x} \, dx$$

$$u = x \quad dv = e^{3x}$$

$$du = dx \quad v = \frac{1}{3} e^{3x}$$

$$\frac{x e^{3x}}{3} - \frac{1}{3} \int e^{3x} \, dx$$

$$\frac{x e^{3x}}{3} - \frac{1}{9} e^{3x}$$

$$\frac{x^2 e^{3x}}{3} - \frac{2}{3} \left(\frac{x e^{3x}}{3} - \frac{e^{3x}}{9} \right)$$

Tabular Method $\int x^2 e^{3x} \, dx$

u	dv
+ x ²	e ^{3x}
- 2x	$\frac{1}{3} e^{3x}$
+ 2	$\frac{1}{9} e^{3x}$
- 0	$\frac{1}{27} e^{3x}$

$$\frac{x^2 e^{3x}}{3} - \frac{2x e^{3x}}{9} + \frac{2 e^{3x}}{27}$$

Ex: $\int x^3 \sin(2x) \, dx =$

u	dv
+ x ³	Sin(2x)
- 3x ²	$-\frac{1}{2} \cos(2x)$
+ 6x	$-\frac{1}{4} \sin(2x)$
- 6	$\frac{1}{8} \cos(2x)$
+ 0	$\frac{1}{16} \sin(2x)$

$$-\frac{1}{2} x^3 \cos(2x) + \frac{3x^2}{4} \sin(2x)$$

$$+\frac{3x}{4} \cos(2x) - \frac{6}{16} \sin(2x)$$

Ex: $\int e^x \sin x \, dx =$

$$u = e^x \quad dv = \sin x \, dx$$

$$du = e^x \, dx \quad v = -\cos x$$

$$-e^x \cos x + \int e^x \cos x \, dx = \int e^x \sin x \, dx$$

$$u = e^x \quad dv = \cos x \, dx$$

$$du = e^x \, dx \quad v = \sin x$$

$$-e^x \cos x + (e^x \sin x - \int e^x \sin x \, dx) = \int e^x \sin x \, dx$$

$$-e^x \cos x + e^x \sin x = 2 \int e^x \sin x \, dx$$

$$\int e^x \sin x \, dx = \frac{-e^x \cos x + e^x \sin x}{2} + C$$